Error Analysis for Solving a Linear System by Cholesky Factorization

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Abstract

Following Demmel [Dem81] with some lemmas improved by Rump and Jeannerod [RJ14] and then implemented in libValidSDP by Roux [Rou16], we derive the backward error analysis for solving a s.p.d. linear system $Ax = b$ with fully accurate consideration of underflows. We do this as a guide to what will be mechanized in Rocq, or what will be documentation of what will have been mechanized.

1 Preliminaries

Almost all of this section is quoted directly from Demmel (1981).

Let ϵ be the rounding error of the arithmetic, and λ be the underflow threshold. Suppose, for example, a binary floating point number is represented as $f \cdot 2^e$ where f lies between 1 and 2 with an n bit fraction, and e is an integer exponent in the range $e_{\min} \le e \le e_{\max}$. Then $\epsilon = 2^{-n}$ (the difference between 1 and the next largest number). The underflow threshold is $\lambda = 2^{e_{\min}}$; this is the smallest ... normalized number for G.U. ... Denormalized numbers lie between 0 and $2^{e_{\min}}$ in G.U. arithmetic for which the smallest nonzero number is thus $\mu = \epsilon\lambda$. We will give a more precise model of rounding and underflow errors in the next section.

Let $\bullet$ be one of the operations $(+, -, *, /)$ and let $fl(a \bullet b)$ denote the floating point result of the indicated computation. ... To take underflow into account, we write ([KP79]),

$$fl(a \bullet b) = (a \bullet b)(1 + e) + \eta \qquad \text{unless } a \bullet b \text{ overflows.} \tag{11}$$

In the case of G.U. we have the following constraints on e and η:

(1) $|e| \le \epsilon$ and $|\eta| \le \lambda\epsilon$,

(2) $\eta \times e = 0$

(3) $\eta = 0$ if $\bullet$ is either addition or subtraction.

We define the function $UN(x)$ to be

$$UN(x) = \begin{cases} 1 & \text{if } x \text{ underflows } (x < \lambda) \\ 0 & \text{if it does not} \end{cases}$$

$\|A\|$ and $\|b\|$ denote the norms of matrix A and vector b. $\|A\|_\infty$ denotes the infinity norm of A, and similarly for $\|b\|_\infty$. $k(A) = \|A\|_\infty \|A^{-1}\|_\infty$ denotes the condition number of the matrix A.

$|A|$ ($|b|$) denotes the matrix (vector) whose entries are the absolute values of the corresponding entries in A (b). Inequalities like $|A| > |B|$ ($|a| > |b|$) are meant componentwise.

Approach. We use backward error analysis. Thus, when trying to solve

$$Ax = b \tag{12}$$

for x we obtain an approximation $\hat{x} = x\ \delta x$ which satisfies the perturbed problem

$$(A + \delta A)\hat{x} = b + \delta b\ . \tag{13}$$

The task of backwards error analysis is to obtain bounds on δA and δb. These bounds can be used in turn to bound the residual r:

$$r = A\hat{x} - b = -\delta A\hat{x} + \delta b \tag{14}$$

2 Comparison with Demmel (1981)

Demmel's Proposition 2 is as follows. As we explain in the next sections, we'll use something slightly different.

To analyze solving a triangular system, we need another expression for the error in computing an inner product:

$$\mathit{float}(\sum_{i=1}^{n} a_i b_i) = \sum_{i=1}^{n} a_i b_i (1 + E_i) + \eta \; . \tag{demmel.31}$$

In the absence of underflow we have

$$\begin{array}{l} |E_1| \le n\epsilon \\ |E_j| \le (n+2-j)\epsilon \quad \text{if} \quad j > 1 \; . \end{array} \tag{demmel.32}$$

In the case of G.U.[1] we have the same bounds on the $|E_i|$, and

$$|\eta| \le n\lambda\epsilon \tag{demmel.34}$$

Demmel's Lemma 4: Error Analysis of Solving a Triangular System. *As we explain in the next sections, we will use a variation of this lemma.*

Consider the lower triangular system $Ly = b$ where L does not necessarily have ones on the diagonal. (The analyses when only ones are on the diagonal or L is upper triangular are similar.) Then in solving $Ly = b$ using the program in Appendix 2 (where inner products are accumulated with roundoff error ϵ_s and smallest number μ_s we obtain $\hat{y} = y + \delta y$ which actually satisfies $(L + \delta L)\hat{y} = b + \delta b$, where

$$|\delta L_{ij}| \le |L_{ij}| \cdot \begin{cases} \epsilon & \text{if} \quad i = j = 1 \\ (i-1)\epsilon_\mathrm{s} & \text{if} \quad i > j = 1 \\ (i+1-j)\epsilon_\mathrm{s} & \text{if} \quad i > j > 1 \\ \epsilon + \epsilon_\mathrm{s} & \text{if} \quad i = j > 1 \end{cases} \tag{demmel.80}$$

independent of underflow.

In the absence of underflow

$$\delta b = 0 \; . \tag{demmel.81}$$

Using G.U.

$$\begin{array}{lllll} \text{if} & i = 1 & |\delta b_i| & \le \lambda\epsilon |L_{11}| UN(y_1) & \text{if } y_1 \text{ underflows} \\ \text{if} & i > 1 & |\delta b_i| & \le i \cdot \lambda_\mathrm{s}\epsilon_\mathrm{s} + & \text{intermediate underflows} \\ & & & \lambda\epsilon |L_{ii}| UN(y_i) & \text{if } y_i \text{ underflows} \end{array} \; . \tag{demmel.82}$$

If L_{ii} is known to be 1, the only changes in the above bounds (besides substituting 1 for L_{ii}) are that $E_{11} = \delta b_1 = 0$ independent of underflow.

3 Error analysis of Cholesky Solve

Instead of Demmel's Proposition 2, we will be using the following lemma:

Rump/Jeannerod Lemma 2.1 (LibValidSDP lemma_2_1_aux): Analysis of $(c - \sum a_i b_i)/b_k$. Let $\hat{y} = \mathit{fl}((c - \sum_{0 \le i < k} a_i b_i)/b_k)$ computed by repeated subtraction left to right. Assume $b_k \ne 0$. Then

$$|b_k \hat{y} - (c - \sum_{0 \le i < k} a_i b_i)| \le (k+1)\epsilon \left(|b_k \hat{y}| + \sum_{0 \le i < k} |a_i b_i| \right) + (1 + (k+1)\epsilon)(k + |b_k|)\lambda\epsilon. \tag{lemma2.1}$$

Demmel's Lemma 4 provides fine-tuned bounds for four different cases, depending on whether $i > j$ and whether $j > 0$. In all these cases we will prove a weaker bound because that allows us to re-use the Rump/Jeannerod Lemma 2.1 for $c - \sum x_i y_i$.

[1] Editor's note: G.U. = Gradual Underflow, that is, a floating-point format that supports subnormal numbers. The technical report also covers the case of "S.Z." or "store zero", which we omit here in all lemmas and theorems.

Theorem Demmel_L4_adapted: Solving the lower triangular system $Ly = b$ we obtain $\hat{y} = y + \delta y$ which satisfies $(L + \delta L)\hat{y} = b + \delta b$, where

$$\begin{array}{rcl} |\delta L_{ij}| & \leq & n\epsilon |L_{ij}| \qquad \text{and} \\ |\delta b_i| & \leq & (1 + n\epsilon)\lambda\epsilon\,(n + |L_{ii}|) \ . \end{array} \qquad \text{(demmel\_lemma\_4)}$$

The factor $(1 + n\epsilon)$ was omitted from Demmel's original but is required for correctness.

Proof. From **??**, we get

$$\left| L_{ii}\hat{y}_i - \left(b_i - \sum_{j=0}^{i-1} L_{ij}\hat{y}_j \right) \right| \leq (i+1)\epsilon \left(|L_{ii}\hat{y}_i| + \sum_{j=0}^{i-1} |L_{ij}\hat{y}_j| \right) + (1 + (i+1)\epsilon)(i + |L_{ii}|)\lambda\epsilon \quad (1)$$

that is

$$\left| b_i - \sum_{j=0}^{i} L_{ij}\hat{y}_j \right| \leq (i+1)\epsilon \left(\sum_{j=0}^{i} |L_{ij}\hat{y}_j| \right) + (1 + (i+1)\epsilon)(i + |L_{ii}|)\lambda\epsilon \quad (2)$$

and since $i < n$, we have $i + 1 \leq n$ which enables to conclude. Q.E.D.

4 Solving a linear system by Cholesky+Substitution

We now derive mixed error bounds for the floating-point result $\hat{x}$ of solving $Ax = b$ by Cholesky decomposition $A = R^T R$ followed by forward substitution $R^T y = b$ then back substitution $Rx = y$.

Lemma th_2_3_aux3: Accuracy of Cholesky decomposition. Suppose A is an $n \times n$ symmetric positive definite floating-point matrix, with $(n+1)\epsilon < 1$. Let $\hat{R}$ be the result of floating-point Cholesky decomposition (inner-product method) of A into $\hat{R}^T\hat{R}$. Let $\delta A = A - \hat{R}^T\hat{R}$. Let the vector $\hat{R}_j$ be the jth column of $\hat{R}$. Let $a_{\max}$ be the maximum diagonal element of $|A|$. Then

$$|\delta A_{ij}| < (\min(i,j) + 2)\epsilon \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 + 4\lambda\epsilon(n + 1 + a_{\max}). \qquad \text{(2.3aux3)}$$

Proof. See Rump+Jeannerod, or see libValidSDP. Q.E.D.

Definition. Define $a'_{\max}$ as $\frac{a_{\max} + 2(n-1)\lambda\epsilon}{1-(n+1)\epsilon}$.

Remark. Let A and $\hat{R}$ be as above. Then

$$\|\hat{R}_j\|_2 \leq \sqrt{a'_{\max}}. \qquad \text{(2.3aux1)}$$

Proof. LibValidSDP th_2_3_aux1 gives

$$\|\hat{R}_j\|_2 \leq \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1-(j+2)\epsilon)}. \quad (3)$$

and the definition of $a'_{\max}$ enables to conclude. Q.E.D.

Corollary. Combining (2.3aux3) and (2.3aux1), we have,

$$\begin{array}{rl} |\delta A_{ij}| & < (\min(i,j) + 2)\epsilon\sqrt{a'_{\max}} \cdot \sqrt{a'_{\max}} + 4\lambda\epsilon(n + 1 + a_{\max}) \\ & \leq (n+1)\epsilon a'_{\max} + 4\lambda\epsilon(n + 1 + a_{\max}) \end{array} \qquad \text{(ch\_back\_err)}$$

Lemma. We obtain bounds on $\delta R^T\hat{R}$ by combining (2.3aux1) and (demmel_lemma_4):

$$\begin{array}{rll} |(\delta R^T\hat{R})_{ij}| & \leq (|\delta R^T| \cdot |\hat{R}|)_{ij} & \\ & \leq \sum_k n\epsilon |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| & \text{(by demmel\_lemma\_4)} \\ & \leq n\epsilon \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 & \text{(by Cauchy} - \text{Schwartz)} \\ & \leq n\epsilon\sqrt{a'_{\max}}\sqrt{a'_{\max}} & \text{(by2.3aux1)} \\ & = n\epsilon a'_{\max}. & \end{array} \qquad \text{(csolve\_back\_err\_aux1)}$$

The same bound is obtained on $\hat{R}\delta R$.

Lemma. Bounds on $\delta R^T \delta R$ as follows:

$$\begin{aligned}
|(\delta R^T \delta R)_{ij}| &\le (|\delta R^T| \cdot |\delta R|)_{ij} \\
&\le \sum_k |\delta R^T|_{ik} |\delta R|_{kj} \\
&\le \sum_k n^2 \epsilon^2 |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| \\
&\le n^2 \epsilon^2 \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 \\
&\le n^2 \epsilon^2 a'_{\max}.
\end{aligned} \qquad \text{(csolve_back_err_aux2)}$$

Lemma. We obtain bounds on $\delta b'$ by combining (demmel_lemma_4) and (2.3aux1):

$$\begin{aligned}
|\delta b'_i| &\le (1+n\epsilon)\lambda\epsilon(n + |\hat{R}^T_{ii}|) \\
&\le (1+n\epsilon)\lambda\epsilon(n + \|\hat{R}_i\|_2) \\
&\le (1+n\epsilon)\lambda\epsilon(n + \sqrt{a'_{\max}}) \\
&\le (1+n\epsilon)^2\lambda\epsilon(n + \sqrt{a'_{\max}}).
\end{aligned} \qquad \text{(csolve_back_err_aux3)}$$

Lemma. We obtain bounds on $(\hat{R}^T + \delta R^T)\delta y$ by combining (demmel_lemma_4) and (2.3aux1):

$$\begin{aligned}
|((\hat{R}^T + \delta R^T)\delta y)_i| &\le (|\hat{R}^T + \delta R^T| \cdot |\delta y|)_i \\
&\le \sum_k |\hat{R}^T + \delta R^T|_{ik} |\delta y|_k \\
&\le \sum_k (1+n\epsilon)|\hat{R}_{ki}| \cdot |\delta y|_k \qquad \text{(by demmel_lemma_4)} \\
&\le (1+n\epsilon)\|\hat{R}_i\|_2 \|\delta y\|_2 \qquad \text{(by Cauchy} - \text{Schwartz)} \\
&\le (1+n\epsilon)\sqrt{a'_{\max}} \|\delta y\|_2 \qquad \text{(by 2.3aux1)} \\
&\le (1+n\epsilon)\sqrt{a'_{\max}}\, n \|\delta y\|_\infty \\
&\le (1+n\epsilon)\sqrt{a'_{\max}}\, n(1+n\epsilon)\lambda\epsilon(n + \sqrt{a'_{\max}}) \\
&= (1+n\epsilon)\lambda\epsilon(n + \sqrt{a'_{\max}})(1+n\epsilon) n \sqrt{a'_{\max}} \\
&\le (1+n\epsilon)^2\lambda\epsilon(n + \sqrt{a'_{\max}}) n \sqrt{a'_{\max}}.
\end{aligned}$$

(csolve_back_err_aux4)

Theorem: Cholesky factor + substitution. Suppose A is an $n \times n$ symmetric positive definite floating-point matrix , with $(n+1)\epsilon < 1$. Suppose b is a floating-point vector of length n. Suppose $\hat{x}$ is the result of floating-point Cholesky factorization of A into $\hat{R}^T\hat{R}$ followed by forward substitution finding $\hat{y}$ such that $\hat{R}^T\hat{y} = b$ and then $\hat{R}\hat{x} = \hat{y}$. Taking into account underflows in both the factorization and the substitution, we have that there is a mixed error δA and δb such that $(A + \delta A)\hat{x} = b + \delta b$, bounded as follows:

$$\begin{aligned}
|\delta A| &\le ((3n+1)\epsilon + n^2\epsilon^2)a'_{\max} + 4\lambda\epsilon(n + 1 + a_{\max}). \\
|\delta b| &\le (1+n\epsilon)^2(n+1)^2\lambda\epsilon(1 + a'_{\max})
\end{aligned} \qquad \text{(csolve_back_err)}$$

Proof. Since $\hat{y} + \delta y = (\hat{R} + \delta R)\hat{x}$, we have

$$\begin{aligned}
b + \delta b' &= (\hat{R}^T + \delta R^T)\hat{y} \\
b + \delta b' + (\hat{R}^T + \delta R^T)\delta y &= (\hat{R}^T + \delta R^T)(\hat{R} + \delta R)\hat{x} \\
&= (\hat{R}^T\hat{R} + \delta R^T\hat{R} + \hat{R}^T\delta R + \delta R^T\delta R)\hat{x} \\
&= (A + \delta A' + \delta R^T\hat{R} + \hat{R}^T\delta R + \delta R^T\delta R)\hat{x} \\
&= A\hat{x} + (\delta A' + \delta R^T\hat{R} + \hat{R}^T\delta R + \delta R^T\delta R)\hat{x}.
\end{aligned} \qquad \text{(xxx)}$$

We will define

$$\begin{aligned}
\delta A &= \delta A' + \delta R^T\hat{R} + \hat{R}^T\delta R + \delta R^T\delta R \\
&\le |\delta A'| + |\delta R^T\hat{R}| + |\hat{R}^T\delta R| + |\delta R^T\delta R|
\end{aligned} \qquad \text{(xxx)}$$

and

$$\delta b = \delta b' + (\hat{R}^T + \delta R^T)\delta y. \qquad \text{(xxx)}$$

From (ch_back_err) we have $|\delta A'_{ij}| < (n+1)\epsilon a'_{\max} + 4\lambda\epsilon(n + 1 + a_{\max})$.
From csolve_back_err_aux1 we have $|(\delta R^T\hat{R})_{ij}| \le n\epsilon a'_{\max}$ and similarly $|(\hat{R}^T\delta R)_{ij}| \le n\epsilon a'_{\max}$.
From csolve_back_err_aux2 we have $|\delta R^T\delta R| \le n^2\epsilon^2 a'_{\max}$.
The result follows using csolve_back_err3 and csolve_back_err4 and $\sqrt{a'_{\max}} \le 1 + a'_{\max}$. Q.E.D.

References

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